

Unit 8: Polar & Parametric Equations

Honors Precalculus

Student

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8.1 Polar Coordinates

Objectives

- Plot points using polar coordinates.
- Convert polar coordinates to rectangular coordinates.
- Convert rectangular coordinates to polar coordinates.
- Transform equations between polar and rectangular forms.

The Polar Coordinate System

The **polar coordinate system** is a two-dimensional coordinate system built from:

- the **pole** — a fixed _____ point.
- the **polar axis** — a fixed ray from the pole pointing in the positive x -direction.

A point P is described by the ordered pair (r, θ) :

- r — the _____ from the pole to P .
- θ — the _____ from the polar axis to the segment OP , measured counterclockwise (positive) or clockwise (negative).

Negative r : If $r < 0$, move in the _____ direction from θ . The point $(-r, \theta)$ is the same as $(r, \theta + \pi)$.

Multiple representations: Every point has infinitely many polar coordinates. All of the following describe the same point:

$$(r, \theta) = (r, \theta + 2\pi k) = (-r, \theta + \pi) \quad \text{for any integer } k.$$

Example 1

Plot the points $(2, 60^\circ)$, $(3, \frac{\pi}{2})$, $(-2, 30^\circ)$, and $(1, \frac{7\pi}{4})$ on a polar grid.

Polar–Rectangular Conversion Formulas

Given a point (r, θ) in polar or (x, y) in rectangular:

Polar $\rightarrow$ Rectangular	Rectangular $\rightarrow$ Polar
$x =$ _____	$r^2 =$ _____
$y =$ _____	$\tan \theta =$ _____

Finding θ : Use $\theta = \arctan(\frac{y}{x})$, then _____ for the correct quadrant based on the signs of x and y .

Example 2

Convert the polar point $\left(3, \frac{\pi}{2}\right)$ to rectangular coordinates.

Example 3

Convert the polar point $\left(-2, \frac{\pi}{6}\right)$ to rectangular coordinates.

Example 4

Convert the rectangular point $(3, 3)$ to polar coordinates.

Example 5

Convert the rectangular point $(-1, \sqrt{3})$ to polar coordinates.

Note**Converting Equations**

Rectangular $\rightarrow$ Polar: Replace x with $r \cos \theta$ and y with $r \sin \theta$. Simplify using $r^2 = x^2 + y^2$ when possible.

Polar $\rightarrow$ Rectangular: Multiply both sides by r if needed to create r^2 , $r \cos \theta$, or $r \sin \theta$, then substitute.

Example 6

Convert $x^2 + y^2 = 9$ to a polar equation.

Example 7

Convert the polar equation $r = 2 \sec \theta$ to rectangular form.

Example 8

Convert $r = 3 \sin \theta$ to rectangular form and identify the curve.

Fact**Standard Line Forms in Polar:**

Vertical line $x = c$: _____

Horizontal line $y = c$: _____

Homework: OpenStax Section 8.3 Exercises (odds recommended)

8.2 Polar Coordinates: Graphs

Objectives

- Test polar equations for symmetry.
- Graph polar equations by plotting points.
- Identify and sketch standard polar curve families.

Symmetry Tests for Polar Graphs

Important: These tests are *sufficient* but not *necessary*. A curve can be symmetric even if the test is inconclusive.

Symmetry	Substitute	Result
About $\theta = \frac{\pi}{2}$ (y-axis)	Replace (r, θ) with _____	Equation unchanged $\Rightarrow$ symmetric
About polar axis (x-axis)	Replace (r, θ) with _____	Equation unchanged $\Rightarrow$ symmetric
About the pole (origin)	Replace (r, θ) with _____	Equation unchanged $\Rightarrow$ symmetric

Example 1

Test $r = 2 \sin \theta$ for symmetry about $\theta = \frac{\pi}{2}$.

Note**Graphing Procedure:**

1. Test for symmetry to reduce the number of points needed.
2. Find zeros: set $r = 0$ and solve for θ .
3. Find the maximum $|r|$.
4. Build a t-table: choose key values of θ (multiples of $\frac{\pi}{6}$ or $\frac{\pi}{4}$), compute r .
5. Plot (r, θ) pairs. Use symmetry to fill in remaining portions.

Example 2

Graph $r = 2 \sin \theta$. Find the zeros, maximum, and sketch the curve.

Standard Polar Curve Families

Family	Equation	Key Feature
Circle	$r = a \cos \theta$ or $r = a \sin \theta$	Radius _____; center on an axis
Cardioid	$r = a \pm b \cos \theta$ or $\sin \theta$, $\frac{a}{b} = \text{_____}$	_____ -shaped; passes through pole
Limaçon (dimpled)	same form, _____	Dimple, no inner loop
Limaçon (inner loop)	same form, _____	Inner loop at pole
Lemniscate	$r^2 = a^2 \cos 2\theta$ or $a^2 \sin 2\theta$	_____; $\max r = a $
Rose curve	$r = a \cos n\theta$ or $a \sin n\theta$	n odd $\rightarrow$ _____ petals; n even $\rightarrow$ _____ petals
Archimedes' spiral	$r = \theta$, $\theta \geq 0$	Outward spiral; no formal symmetry

Example 3

Graph the circle $r = 4 \cos \theta$. Find the zeros and maximum.

Example 4

Identify the family and sketch $r = 2 + 2 \cos \theta$.

Example 5

Identify the family and sketch $r = 3 + 2 \cos \theta$.

Example 6

Identify the family and sketch $r = 1 + 2 \cos \theta$. Note any values of θ where $r < 0$.

Example 7

Identify the family and sketch $r^2 = 4 \cos 2\theta$.

Example 8

Identify the family and state the petal count for $r = 2 \cos 4\theta$.

Archimedes' Spiral

The spiral $r = \theta$ ($\theta \geq 0$) passes through the pole at $\theta = 0$ and spirals outward. There is no formal algebraic symmetry. Build a t-table using $\theta = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$.

Homework: OpenStax Section 8.4 Exercises (odds recommended)

8.3 Polar Form of Complex Numbers**Objectives**

- Plot complex numbers in the complex plane and find their absolute value.
- Write complex numbers in polar form.
- Convert between polar and rectangular forms.
- Multiply and divide complex numbers using polar form.
- Apply De Moivre's Theorem to find powers of complex numbers.

Note

Connection to Unit 2.5: You already know that a complex number $z = a + bi$ can be plotted at the point (a, b) on the complex plane, where the horizontal axis is _____ and the vertical axis is _____. Today we add a second way to describe the same point: distance and angle — exactly the polar coordinates from Section 8.1.

Absolute Value (Modulus) of a Complex Number

For $z = a + bi$, the **absolute value** (or **modulus**) is:

$$|z| = \underline{\hspace{2cm}}$$

Geometrically, $|z|$ is the _____ from the origin to z in the complex plane.

Example 1

Find $|\sqrt{5} - i|$.

Example 2

Find $|3 - 4i|$.

Polar Form of a Complex Number

Every complex number $z = a + bi$ can be written as:

$$z = \underline{\hspace{2cm}}$$

where

- $r = |z| = \underline{\hspace{2cm}}$ is the **modulus**.
- θ is the **argument**: $\tan \theta = \underline{\hspace{2cm}}$ (check quadrant).

Shorthand: $r(\cos \theta + i \sin \theta) = \underline{\hspace{2cm}}$

Rectangular from polar: $a = r \cos \theta$ and $b = r \sin \theta$.

Example 3

Write $4i$ in polar form.

Example 4

Write $-4 + 4i$ in polar form.

Example 5

Convert $z = 12(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$ to rectangular form.

Fact**Multiplication and Division in Polar Form**

If $z_1 = r_1 \operatorname{cis} \theta_1$ and $z_2 = r_2 \operatorname{cis} \theta_2$:

Product: $z_1 z_2 =$ _____ (multiply moduli, _____ arguments)

Quotient: $\frac{z_1}{z_2} =$ _____ (divide moduli, _____ arguments)

Example 6

Find $z_1 z_2$ where $z_1 = 4 \operatorname{cis}(80^\circ)$ and $z_2 = 2 \operatorname{cis}(145^\circ)$.

Example 7

Find $\frac{z_1}{z_2}$ where $z_1 = 2 \operatorname{cis}(213^\circ)$ and $z_2 = 4 \operatorname{cis}(33^\circ)$.

De Moivre's Theorem and nth Roots

De Moivre's Theorem: For $z = r \operatorname{cis} \theta$ and positive integer n :

$$z^n = \underline{\hspace{2cm}}$$

nth Root Theorem: The n distinct n th roots of $z = r \operatorname{cis} \theta$ are:

$$z^{1/n} = r^{1/n} \operatorname{cis} \left(\frac{\theta}{n} + \frac{2\pi k}{n} \right), \quad k = 0, 1, \dots, n-1$$

The roots are equally spaced on a circle of radius $r^{1/n}$.

Example: Evaluate $(1 + i)^5$.

Example: Find all cube roots of $z = 8 \operatorname{cis}\left(\frac{2\pi}{3}\right)$.

Homework: OpenStax Section 8.5 Exercises (odds recommended)

8.4 Parametric Equations

Objectives

- Parameterize a curve by writing x and y as functions of a parameter.
- Eliminate the parameter to find the corresponding Cartesian equation.
- Model linear motion with parametric equations.

Parametric Equations

A pair of equations

$$x = f(t), \quad y = g(t)$$

defines a **plane curve** as t varies. The variable t is the **parameter**.

- Each value of t gives an ordered pair $(x(t), y(t))$ on the curve.
- As t increases, the curve is traced in a specific _____ (called the **orientation**).
- The parameter t often represents _____, but it can represent any indexing quantity (angle, arc length, etc.).

Key difference from $y = f(x)$: A parametric curve carries direction information. Two different sets of parametric equations can produce the same set of (x, y) points but trace them in _____ directions — they are different curves.

Example 1

Parameterize $y = x^2 - 1$. Build a t-table and sketch the curve. Indicate direction of travel with arrows.

Example 2

An object travels at a constant rate from $(-5, 3)$ to $(3, -1)$ in 4 seconds. Write parametric equations for its position at time t .

Eliminating the Parameter

Goal: Convert the parametric pair to a single equation in x and y .

General strategy:

1. Identify which equation is easier to _____.
2. Solve for t and _____ into the other equation.
3. Simplify to find the relationship between x and y only.
4. **Check the domain:** the Cartesian equation may have a _____ domain than the parametric curve — always verify.

Trigonometric method (for $x = a \cos t$, $y = b \sin t$): Solve for $\cos t$ and $\sin t$, then apply $\cos^2 t + \sin^2 t =$ _____.

Example 3

Eliminate the parameter: $x(t) = t^2 + 1$, $y(t) = 2 + t$. Identify the curve.

Example 4

Eliminate the parameter: $x(t) = e^{-t}$, $y(t) = 3e^t$. Identify the curve and note any domain restrictions.

Example 5

Eliminate the parameter: $x(t) = \sqrt{t} + 2$, $y(t) = \log t$. State the domain.

Example 6

Eliminate the parameter from $x(t) = 4 \cos t$, $y(t) = 3 \sin t$, $0 \leq t \leq 2\pi$. Identify the curve.

Fact

What Parametric Form Tells You That $y = f(x)$ Cannot:

- The _____ in which the curve is traced (orientation).
- The _____ of the object at a specific time t .
- Whether a self-intersecting curve crosses at different _____.

Homework: OpenStax Section 8.6 Exercises (odds recommended)

8.5 Parametric Equations: Graphs

Objectives

- Graph parametric equations by plotting points from a t-table.
- Indicate the direction of travel (orientation) with arrows.
- Apply parametric equations to projectile motion problems.

Note**Graphing a Parametric Curve — Step by Step:**

1. Build a t -table with columns t , $x(t)$, $y(t)$.
2. Evaluate at evenly-spaced values of t across the given interval.
3. Plot each ordered pair (x, y) .
4. Connect the points with a smooth curve.
5. Add _____ showing the path as t increases.

Desmos tip: In parametric mode, use the t -slider to animate the curve and see direction of travel in real time.

Example 1

Graph $x(t) = t^2 + 1$, $y(t) = 2 + t$ for $-3 \leq t \leq 3$. Indicate direction of travel.

Example 2

Graph $x = 2 \cos t$, $y = 4 \sin t$ for $0 \leq t \leq 2\pi$. Identify the curve and indicate direction.

Projectile Motion Equations

For an object launched at speed v_0 (ft/s or m/s), angle θ above horizontal, from initial height h :

Horizontal: $x(t) = \underline{\hspace{2cm}}$

Vertical: $y(t) = \underline{\hspace{2cm}}$

where $g = \underline{\hspace{2cm}}$ ft/s² (US customary) or $\underline{\hspace{2cm}}$ m/s² (metric).

Note: In the vertical equation, the t^2 coefficient is $\frac{1}{2}g$, so it is _____ (not 32) in the ft/s² formula.

Example 3

A baseball is hit with an initial speed of 140 ft/s at an angle of 45° from a height of 3 feet.

- (a) Write the parametric equations for the ball's position.
- (b) Where is the ball at $t = 2$ seconds?
- (c) When does the ball land? How far does it travel?
- (d) A 10-foot fence stands at $x = 400$ feet. Does the ball clear it?

Changing Launch Angle

Redo Example 3 with $\theta = 30^\circ$. How does the trajectory change? Does the ball still clear a fence at $x = 400$ ft?

Homework: OpenStax Section 8.7 Exercises (odds recommended)

8.6 Vectors

Objectives

- Write vectors in component form and in $\mathbf{i}, \mathbf{j}$ notation.
- Find the magnitude and direction of a vector.
- Perform vector addition, subtraction, and scalar multiplication.
- Find the unit vector in a given direction.
- Compute the dot product and find the angle between two vectors.

Vectors and Scalars

A **scalar** is a quantity described by _____ only (a number).

A **vector** is a quantity described by both _____ and _____. It is represented geometrically as an arrow.

Component form: A vector with initial point at the origin and terminal point (a, b) is written:

$$\mathbf{v} = \underline{\hspace{2cm}}$$

The values a and b are the **components** of $\mathbf{v}$.

From any two points: For initial point (x_1, y_1) and terminal point (x_2, y_2) :

$$\mathbf{v} = \underline{\hspace{2cm}}$$

Example 1

Find the component form of the vector with initial point $P(2, 3)$ and terminal point $Q(6, 4)$.

Magnitude and Direction

For $\mathbf{v} = \langle a, b \rangle$:

Magnitude: $|\mathbf{v}| = \underline{\hspace{2cm}}$

Direction angle θ : $\tan \theta = \underline{\hspace{2cm}}$ (check quadrant!)

Components from magnitude and angle: $a = \underline{\hspace{2cm}}$, $b = \underline{\hspace{2cm}}$

Example 2

Find the magnitude and direction angle of $\mathbf{v} = \langle 3, 4 \rangle$.

Example 3

A vector has magnitude 7 and direction angle 30° . Write it in component form.

Vector Operations

Let $\mathbf{u} = \langle a, b \rangle$ and $\mathbf{v} = \langle c, d \rangle$, and let k be a scalar.

Operation	Formula	Geometric meaning
Addition	$\mathbf{u} + \mathbf{v} = \underline{\hspace{2cm}}$	Tip-to-tail of arrows
Subtraction	$\mathbf{u} - \mathbf{v} = \underline{\hspace{2cm}}$	Reverse $\mathbf{v}$, then add
Scalar multiplication	$k\mathbf{v} = \underline{\hspace{2cm}}$	Scales magnitude; reverses direction if $k < 0$

Example 4

Let $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle -1, 4 \rangle$. Find (a) $\mathbf{u} + \mathbf{v}$, (b) $\mathbf{u} - \mathbf{v}$, (c) $3\mathbf{v}$.

Standard Unit Vectors and $\mathbf{i}$ / $\mathbf{j}$ Notation

The **standard unit vectors** are:

$$\mathbf{i} = \langle 1, 0 \rangle \quad \mathbf{j} = \langle 0, 1 \rangle$$

Any vector $\langle a, b \rangle$ can be written as:

$$\mathbf{v} = \underline{\hspace{2cm}}$$

A **unit vector** in the direction of $\mathbf{v}$ is:

$$\hat{\mathbf{v}} = \underline{\hspace{2cm}} = \left\langle \frac{a}{|\mathbf{v}|}, \frac{b}{|\mathbf{v}|} \right\rangle$$

It always has magnitude $\underline{\hspace{1cm}}$.

Example 5

Find the unit vector in the direction of $\mathbf{v} = \langle -5, 12 \rangle$. Write the result in component form and in $\mathbf{i}, \mathbf{j}$ notation.

Dot Product and Angle Between Vectors

Dot product of $\mathbf{v} = \langle a, b \rangle$ and $\mathbf{u} = \langle c, d \rangle$:

$$\mathbf{v} \cdot \mathbf{u} = \underline{\hspace{2cm}}$$

The result is a _____ (not a vector).

Angle between two vectors:

$$\cos \theta = \underline{\hspace{2cm}}$$

Perpendicular vectors: $\mathbf{v} \perp \mathbf{u}$ if and only if $\mathbf{v} \cdot \mathbf{u} = \underline{\hspace{2cm}}$.

Example: Find $\mathbf{v} \cdot \mathbf{u}$ and the angle between $\mathbf{v} = \langle 5, 12 \rangle$ and $\mathbf{u} = \langle -3, 4 \rangle$.

Example 6

An airplane flies at 200 mph on a compass bearing of 140° (measured clockwise from north). A wind blows due north at 16.2 mph. Find the airplane's actual ground speed and bearing.

Homework: OpenStax Section 8.8 Exercises (odds recommended)